

Program 6: 3×3 Transformation Matrix using OpenCV

Source Code:

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Read image
img = cv2.imread("/content/images (2).jfif")


if img is None:
    print("Image not found!")
else:
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)


h, w = img.shape[:2]


# Four corner points of original image
pts1 = np.float32([
    [0, 0],
    [w-1, 0],
    [0, h-1],
    [w-1, h-1]
])
```

```
# New corner positions (transformed)
pts2 = np.float32([
    [50, 50],
    [w-100, 20],
    [80, h-50],
    [w-50, h-80]
])

# Compute 3×3 Perspective Transformation Matrix
matrix = cv2.getPerspectiveTransform(pts1, pts2)

# Apply transformation
transformed = cv2.warpPerspective(img, matrix, (w, h))

# Display Original and Transformed images
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(transformed)
```

```
plt.title("3×3 Matrix Transformation")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Print 3×3 matrix
```

```
print("Transformation Matrix:")
```

```
print(matrix)
```

Output:

...

Original Image



3×3 Matrix Transformation



```
Transformation Matrix:  
[[ 6.61732553e-01  5.60999622e-02  5.00000000e+01]  
 [-6.75058840e-02  7.22622520e-01  5.00000000e+01]  
 [-1.20655026e-05 -1.39557647e-04  1.00000000e+00]]
```

Program 7: Image Stretching (Scaling) using OpenCV

Source Code:

```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread("/content/images (2).jfif")

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert BGR to RGB
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Stretch image (2x width, 1.5x height)
stretched = cv2.resize(img, None, fx=2, fy=1.5)
```

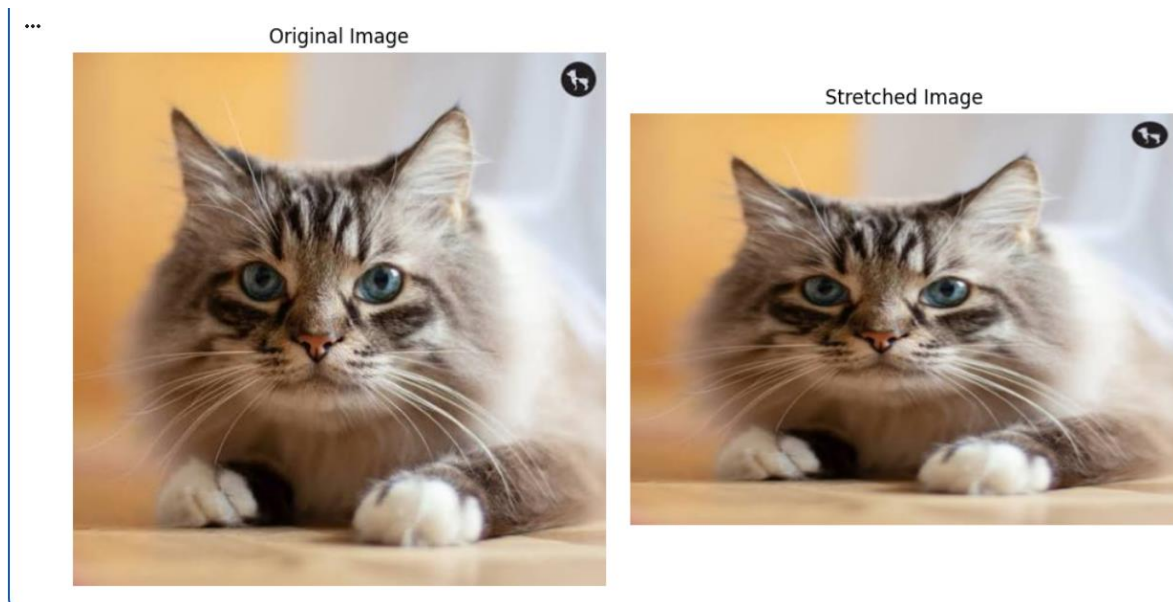
```
# Display Original and Stretched images
plt.figure(figsize=(10,5))
```

```
# Original Image
plt.subplot(1,2,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")
```

```
# Stretched Image
plt.subplot(1,2,2)
plt.imshow(stretched)
plt.title("Stretched Image")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

Output:



Program 7: 90° Image Rotation using OpenCV

Source Code:

```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread("/content/images (2).jfif")

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert BGR to RGB
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Image size
h, w = img.shape[:2]

# Center of image
center = (w // 2, h // 2)

# Rotation matrix (90 degrees)
matrix = cv2.getRotationMatrix2D(center, 90, 1)

# Rotate image
rotated = cv2.warpAffine(img, matrix, (w, h))

# Display Original and Rotated images
plt.figure(figsize=(10,5))

# Original Image
plt.subplot(1,2,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

# Rotated Image
plt.subplot(1,2,2)
plt.imshow(rotated)
plt.title("90° Rotated Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

...

Original Image



90° Rotated Image

